CSU 07405 and ITU 07406: Algorithm Analysis and Design

GROUP ASSIGNMENT (HDCSII and HDIT II)

Five (5) Students ONLY per Group

QUESTION ONE

(i) Write an Algorithm for quick Sort and write its time complexity

(ii) Apply quick sort to the list 5, 3, 1, 9, 8, 2, 4, 7.

QUESTION TWO

Briefly explain an algorithm used for 8-Queens problem using backtracking

QUESTION THREE

Apply the branch and bound technique to solve the traveller salesperson problem

QUESTION FOUR

Define a Knapsack problem. Discuss how you will solve a Knapsack problem with few steps using a Dynamic Programming Algorithm?

Submission Date: 19/05/2025